INCH-POUND
MIL-PRF-32271/16A
12 JANUARY 2015
SUPERSEDING
MIL-PRF-32271/16
09 JULY 2014

PERFORMANCE SPECIFICATION SHEET

BATTERY, NON-RECHARGEABLE, LITHIUM

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product herein shall consist of this specification and MIL-PRF-32271.

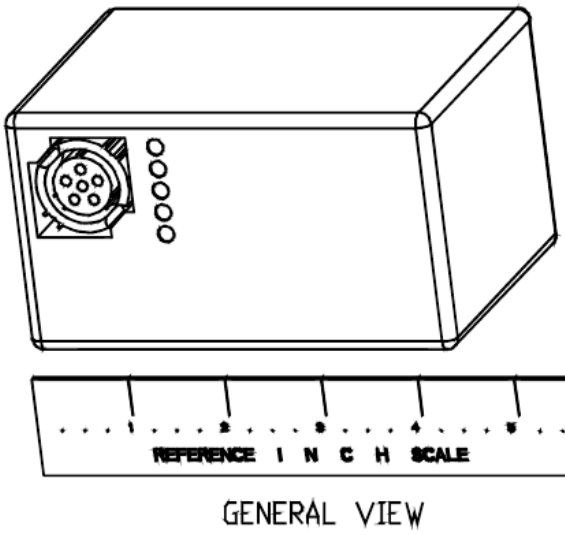


Figure 1 – General View, M32271/16 Battery Shape

REQUIREMENTS (see 1.2 for Type, Class, and Features descriptions):

Type	IV		
Class	2		
Features	C	D	E
PIN	M32271/16-42C	M32271/16-42D	M32271/16-42E

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Specification requirements: The following requirements of MIL-PRF-32271, identified therein by the phrase “when specified”, are applicable as indicated below:

Requirement	Specification Reference	Applicability by PIN Suffix (M32271/16-)		
		42C	42D	42E
Parallel cell arrangements	3.4	No	No	No
Connectors	3.4.5	Yes	Yes	Yes
Battery charger connection	3.4.5.1	No	No	No
Connection integrity	3.4.5.2	No	No	No
Terminal integrity	3.4.6	No	No	No
Complete discharge device	3.4.7	Yes	Yes	Yes
State of charge device	3.4.8	No	Yes	Yes
State of charge data output	3.4.8e	No	No	Yes
Cell charging	3.5.3	Yes	Yes	Yes
Nail penetration	3.5.4	Yes	Yes	Yes
Cell series short circuit	3.5.6	Yes	Yes	Yes
Parallel string charge protection	3.5.6.1	No	No	No
Charge protection	3.5.11	Yes	Yes	Yes
Over-current protection	3.5.13.1	Yes	Yes	Yes
Over-temperature protection	3.5.14	Yes	Yes	Yes
Surface temperature	3.6h	Yes	Yes	Yes
Capacity LR & LRT		No	No	No
Immersion	3.7.8	Yes	Yes	Yes
Watertight integrity	3.7.9	No	No	No
DO NOT CHARGE! marking	3.8.1.2b	Yes	Yes	Yes
Complete discharge device marking/label	3.8.4	Yes	Yes	Yes
State of charge marking	3.8.5	No	Yes	Yes
Simplified battery label	3.8.7	Yes	Yes	Yes

Dimensions, marking and configuration, all types:

Battery – Drawing A3348242
Battery Connector (3.4.5 & 4.6.1.7) – Drawing SC-C-179492[1]
Mating Connector (3.4.5 & 4.6.1.7) – Part Number SM-D-687888-1, Drawing SM-D-687888
Battery Charger Connector (3.4.5.1 & 4.6.1.7.1) – Not Applicable

Battery voltages (3.4.4 & 4.6.1.5), all types:

Battery open-circuit voltage (3.4.4.1 & 4.6.1.5.1):
Maximum – 16.50 volts
Minimum – 14.90 volts

Battery closed circuit voltage (3.4.4.2, 3.6, 4.6.1.5.2, & 4.6.4.1): For capacity & initial voltage delay – 10.0 volts minimum
For closed circuit voltage test - 10.0 volts

Cell closed circuit voltage (3.4.4.3 & 4.6.1.3):
Minimum – 2.0 volts per cell

[1] Certifications for physical properties of connector material required by SC-C-179492

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Maximum weight (3.1 & 4.6.1.6): Type IV – 1.63 lbs. (740 g).
Capacity test requirements (3.6 & 4.6.4), Type IV:

Test	Initial voltage delay Time (MAX) to	Volts	Minimum Capacity		Cut-off Volts
I	1 second	10.00	4.5 hours		10.00
IDC	1 second	10.00	29 hours		10.00
L	60 seconds	10.00	7 hours		10.00
H	1 second	10.00	8.7 hours		10.00
			Storage Period		
			1-Week	4-Week	
IT	1 second	10.00	4.2 hours	4.1 hours	10.00
LT	60 second	10.00	6.5 hours	6.4 hours	10.00
HT	1 second	10.00	8.1 hours	7.9 hours	10.00

Abuse test pulse discharge capacity requirement, all types (3.5.12 & 4.6.2.12i):
Type IV – 8.5 ampere-hours.

METHODS OF EXAMINATION AND TEST:

Verification requirements. The following verification requirements of MIL-PRF-32271, identified therein by the phrase "when specified," are applicable to each battery type as indicated below:

Test Requirement	**Specification Reference**	**Applicability by PIN Suffix (M32271/16-)**		
		42C	42D	42E
Parallel discharges	Tables III, IV, IX, & XI	No	No	No
Connectors	4.6.1.7	Yes	Yes	Yes
Battery charger connection	4.6.1.7.1	No	No	No
Static connection integrity	4.6.1.7.2	No	No	No
Dynamic connection integrity	4.6.1.7.3	No	No	No
Terminal integrity	4.6.1.8	No	No	No
Complete discharge device	4.6.1.9	Yes	Yes	Yes
State of charge device	4.6.1.10	No	Yes	Yes
State of charge data output	4.6.1.10	No	No	Yes
Cell charging	4.6.2.3	Yes	Yes	Yes
Nail penetration	4.6.2.4	Yes	Yes	Yes
Cell series string short	4.6.2.6	Yes	Yes	Yes
Parallel string charge protection	4.6.2.6.1	No	No	No
Charge protection	4.6.2.11	Yes	Yes	Yes
Battery over-current protection	4.6.2.13	Yes	Yes	Yes
Battery over-temperature protection	4.6.2.14	Yes	Yes	Yes
Surface temperature	4.6.4.1	Yes	Yes	Yes
Capacity LR	4.6.4.1.5	No	No	No
Capacity LRT	4.6.4.1.9	No	No	No
Capacity IDC	4.6.4.1.10	Yes	Yes	Yes
Immersion	4.6.5.8	Yes	Yes	Yes
Watertight integrity	4.6.5.9	No	No	No

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Cell closed-circuit voltage test: When cells are tested as specified in 4.6.1.3, load each cell with 2 amperes constant current, 1 ohm constant resistance, or 4 watts constant power.

Battery closed-circuit voltage test: When tested as specified in 4.6.1.5.2, load the battery with 2 amperes constant current, 5 ohms constant resistance, or 20 watts constant power.

Abuse test, pre-discharge: When tested as specified in 4.6.2.12a, discharge with a load of either 2 amperes or 50 watts for 2.25 hours for Type IV batteries.

Abuse test, pulse discharge: When tested as specified in 4.6.2.12i, discharge with a load of 2.5 amperes for 1 minute followed by 1.8 amperes for 4 minutes, cycled continuously to 10.00 volts.

Battery over-current protection: When tested as specified in 4.6.2.13, apply a load to the batteries and subject to a load of either 2 amperes or 48 watts.

Immersion: When tested as specified in 4.6.5.8, apply a load of either 0.4 amperes or 2.3 watts during the storage and immersion time periods specified.

Capacity tests (4.6.4):

Test	Discharge Rate	Duty Cycle
I	2 amperes	Continuous discharge to zero volts, followed by 2-ampere forced discharge for 10 minutes.
L	2 amperes	Continuous discharge to cut-off voltage.
H	2 amperes	Discharge for 5 minutes at 2 amperes, followed by 5 minutes on open circuit; repeat cycle continuously to cut-off voltage.
IT	2 amperes	Continuous discharge to cut-off voltage.
LT	2 amperes	Continuous discharge to cut-off voltage.
HT	2 amperes	Discharge for 5 minutes at 2 amperes, followed by 5 minutes on open circuit; repeat cycle continuously to cut-off voltage.
IDC	4.6/6.0/20W	Discharge for 4.6W for 6 minutes, then 6.0W for 3 minutes, then 20W for 1 minute; repeat cycle until cut-off voltage.

NOTES:
(This section contains information of a general nature which may be helpful, but is not mandatory)

Intended use: This battery is intended to operate portable military communications electronics equipment. It is used mainly in tactical radios, but is also used in sensors, chemical agent alarms, night vision equipment, and many other types of electronic equipment. Some applications use the battery for powering electric motors. This battery can power any system that utilizes the BA-5390, BA-5590, BA-5790, BB-390 or BB2590 in 12 volt mode, but utilizing half the battery footprint. A spacer may be needed if used in this way.

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[6]Nominal ratings: The following are the nominal ratings for each battery type described by this specification sheet. They are provided for information purposes.

Battery PIN:	**M32271/1-42C, 42D or 42E (Li/CFx-MnO2)**
Overall Dimensions:	2.85 in H x 4.4 in W x 2.45 in D
MAX Weight:	1.63 lbs. (740 g)
Voltage Range:	10-16.5
Nominal Capacity (I-test) (Ampere-hours): (IDC-test)	8.7-10.6 14.6
Nominal Energy (Watt-hours):	123-193
MAX Rated Power Output:	25 watts
MAX Continuous load Rating:	2 Amps
MAX Pulse Rating:	2.5 Amps
Instantaneous Trip Rating:	4 Amps
Operating Temperature Range:	-20 to 130ºF (-29 to 55ºC)
Storage Temperature Range:	-40 to 160ºF (-40 to 71ºC)
MAX Abusive Temperature (non-operating):	192ºF (89ºC)

Other data:

Military Type Designations. The military type designations that relate to the PINs covered by this specification sheet are BA-5795()/U for the Type IV PINs.

The battery shall have 2 active pins. Pins 1 and 4 shall be active and display voltage while pins 2 and 5 shall be disconnected.

All battery types are subject to the transportation requirements of 49 CFR 173.185. M32271/16-42C, -42D and -42E batteries (solid cathode) contain more than 1 gram of lithium per cell and more than 2 grams per battery. Check the appropriate Material Safety Data Sheet for actual lithium content. It should be noted that all non-rechargeable lithium batteries are restricted from shipment as cargo aboard passenger aircraft within, entering, or leaving the US.

All battery types described by this specification sheet include a complete discharge device in order to render the battery non-reactive after use. If the battery is damaged, or if the device does not operate as indicated by an amber or yellow light, the battery is considered reactive waste. Any type, whether completely discharged or not, may be recycled after use. Universal waste rules, where applicable, apply only when recycling is the chosen disposal method.

[6] Capacity and energy delivered by a battery are significantly affected by usage conditions, such as temperature and loads applied. If you have any questions about use of this battery in a particular device or circumstance, please visit the following web site (contact info is posted): http://www.cerdec.army.mil/c2d/armypower.

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The US Army CECOM Life Cycle Management Command (LCMC) publishes guidelines for the design of battery compartments for devices using both battery types I and II covered by this specification sheet: Technical Bulletin CECOM-TB-7. Please contact the CECOM LCMC Directorate for Safety at monm-amselsfsec@conus.army.mil for further information.

Navy safety tests of NAVSEA S9310-AQ-SAF-010 are required for this battery during first article testing. The test data provided from the testing will be used to evaluate applications for specific using devices and Navy platforms. Devices using any battery described by this specification sheet will require US Navy Safety Approval prior to use by Department of Navy users unless such approval has already been granted. Please consult with NAVSEA Instruction 9310.1 for further information.

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Army – CR
Navy – SH
Air Force – 99
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Review activities:
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Air Force – 71

Preparing activity:
Army – CR
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